

PROBLEM SET 2

ECON301—ECONOMETRICS I

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1. Suppose you have the following data on 11 student's combined SAT scores X and their cumulative grade point average at graduation Y .¹

Y_i	X_i
3.63	1490
2.37	1300
3.33	1510
3.32	1420
3.27	1490
2.37	1180
3.61	1550
3.23	1460
2.59	1300
3.30	1450
3.21	1550

- (a) Write down the econometric model relating X and Y (i.e. the linear regression model).
(b) Calculate the following:

$$n; \bar{X}; \bar{Y}; \sum (X_i - \bar{X})(Y_i - \bar{Y}); \sum (X_i - \bar{X})^2; \hat{\beta}_0 \text{ and } \hat{\beta}_1.$$

- (c) Calculate $\hat{Y}_i$ and $\hat{u}_i$ for each observation.
(d) Calculate R^2 .
(e) Compute $\sum \hat{u}_i$. Explain why do you get this result.

¹You can use any spreadsheet software to calculate the calculate what is asked in this exercise.

2. Do problems 4, 6 and 12 from Chapter 2 in the book.
3. Suppose that you have a random sample $\{(x_i, y_i) : i = 1, \dots, n\}$. Consider the following problem:

$$\min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 \quad (1)$$

where b_0 and b_1 are dummy arguments for the optimization problems.

Show that the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are the solution to problem (1). Be as rigorous as you can in your proof, i.e., you will have to show that indeed $(\hat{\beta}_0, \hat{\beta}_1)$ achieves a minimum in the objective of problem (1).

4. Suppose that a researcher, using data on class size (CS) and average test scores from 100 third-grade classes, estimates the OLS Regression

$$\widehat{TestScore} = \underset{(20.4)}{520.4} - \underset{(2.21)}{5.82} CS, \quad R^2 = 0.08, RSE = 11.5,$$

where the standard errors for the coefficient estimates are given by the numbers in parentheses.

- (a) Suppose a classroom has 20 students. What is the regression's prediction for that classroom's average test score?
 - (b) Last year a classroom had 19 students, and this year it has 21 students. What is the regression's prediction for the change in the classroom average test score?
 - (c) The sample average class size across the 100 classrooms is 21.4. What is the sample average of the test scores across the 100 classrooms? (*Hint*: Review the formulas for the OLS estimators).
 - (d) What is the sample standard deviation of the test scores across the 100 classrooms? (*Hint*: Review the formulas for the R^2 and RSE).
5. **Optional.** Do the empirical exercise C9 from chapter 2 in the book. You will have to attach a printout of your R code and output.

(i) Below is the necessary code to answer the questions

```
rm(list=ls())
library(wooldridge)
library(tidyverse)

data("countymurders")

num0murder<-countymurders%>%filter(murders==0)%>%nrow()
numatleast1exec<-countymurders%>%filter(execs>=1)%>%nrow()
summary(select(countymurders,execs))
```

There are 15651 counties had zero murders in 1996. There are 205 counties had at least one execution. The largest number of executions is 7.

(ii) The R code to estimate the model is

```
regression5<-lm(murders~execs,data=countymurders)
summary(regression5)
```

The estimated equation is

$$\widehat{murders} = \underset{(0.2418)}{6.8382} - \underset{(2.1463)}{65.4650} execs, \quad R^2 = 0.02431, RSE = 46.64, n = 37349$$

(iii) The coefficient on *execs* is 65.46, which is positive. It means that one more execution in one county will have 65.46 more murders. Thus, it doesn't suggest that capital punishment is a deterrent for murder.

- (iv) When $execs = 0$, the smallest number of murders that can be predicted by the equation is 6.84. The residual for a county with zero executions and zero murders is -6.8382 .
- (v) In this single variable regression, it is highly suspected that the error term is correlated with the factor 'execution', which violates the assumption SLR4 for OLS, i.e. $E(u|execs) \neq 0$. For example, in a poor county it is much likely that it has both a large number of executions and a large number of murders. In order to isolate this correlation, we need add some other control factors, which further need a multiple regression.